

Accident Analytics: Unveiling Trends and Insights from US Road Data

CS5618 - Business Intelligence

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Introduction

- Every year in the United States, many people die, and enormous amounts of money are damaged due to road accidents. Accident analysis proves the causes of accidental occurrences and dangerous areas, with information on policies to prevent probable accidents. This work employs the US Accidents dataset obtained from Kaggle to identify patterns and offer suggestions to reduce road fatalities.
- The project aims to achieve the following objectives:
 - **Analyze Accident Trends**
 - **Identify Key Risk Factors:** Find factors that significantly increase accident severity.
 - **Actionable Recommendations:** Bring research results that would be beneficial for decision making by the policymakers and advocates of road safety.

Why Business Intelligence for Road Safety?

Data-Driven Decision Making

- Transform 3M+ accident records into actionable insights
- Monitoring of accident hotspots and risk factors
- Predictive analytics for proactive safety measures

Economic Impact

- \$242 billion annual cost of motor vehicle crashes
- Reduce insurance claims through targeted interventions
- Optimize emergency response resource allocation

Key Features:

- Accident hotspot mapping
- Weather impact analysis
- Severity prediction model
- Temporal pattern visualization

Key Insights from Analysis

Key Features:

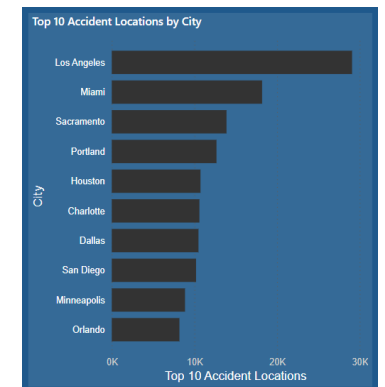
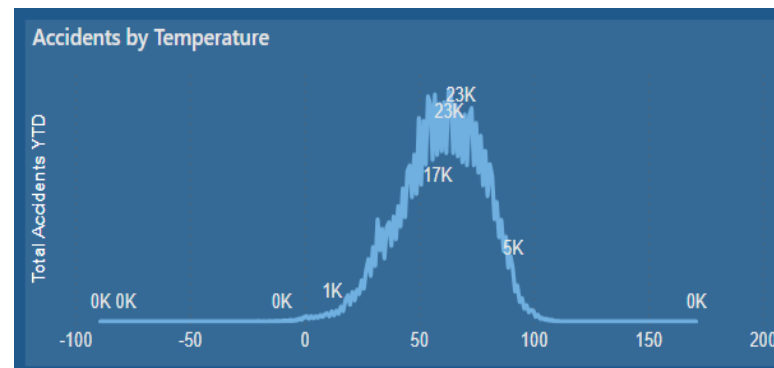
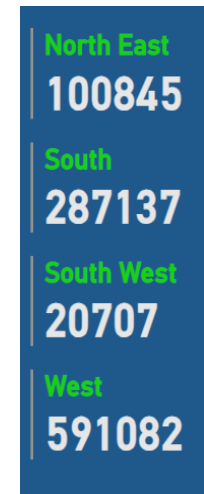
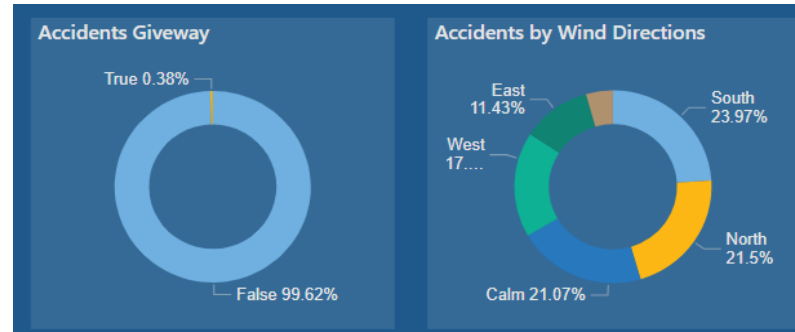
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- Severity prediction model
- Temporal pattern visualization
- Emergency response optimization

Temporal Patterns

- Peak accident hours: 8-9 AM and 4-6 PM (office going and Coming times)
- Higher severity during day hours
- increase in accidents during adverse weather

Key contributing factors:

- Weather conditions (cloudy,rainy,windy)
- Time of day (day/night)
- Road features (traffic signals/bum/junction)



US ROAD ACCIDENT ANALYSIS

Twilight
All

Weather Conditions
All

Total Accidents
999.8K

Low Severe
28.2K
2.82

Medium Severe
771.2K
0.77

High Severe
119.5K
0.12

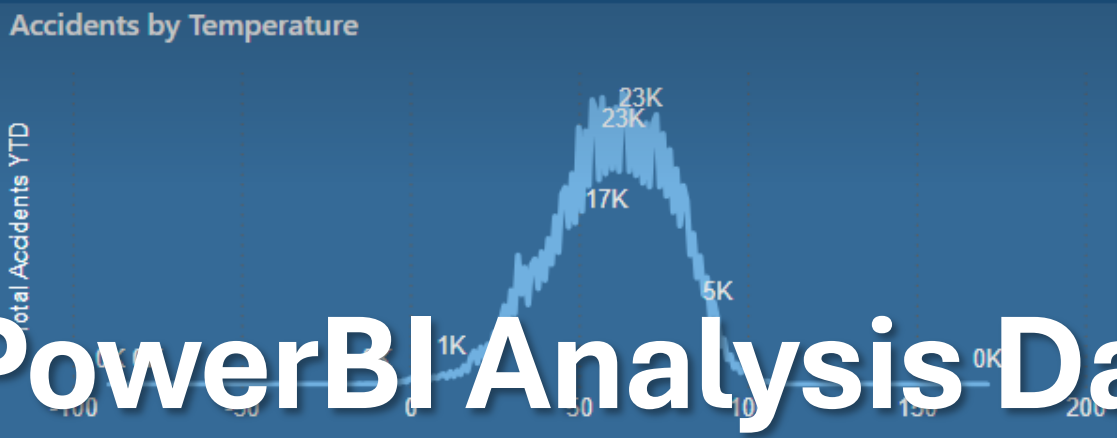
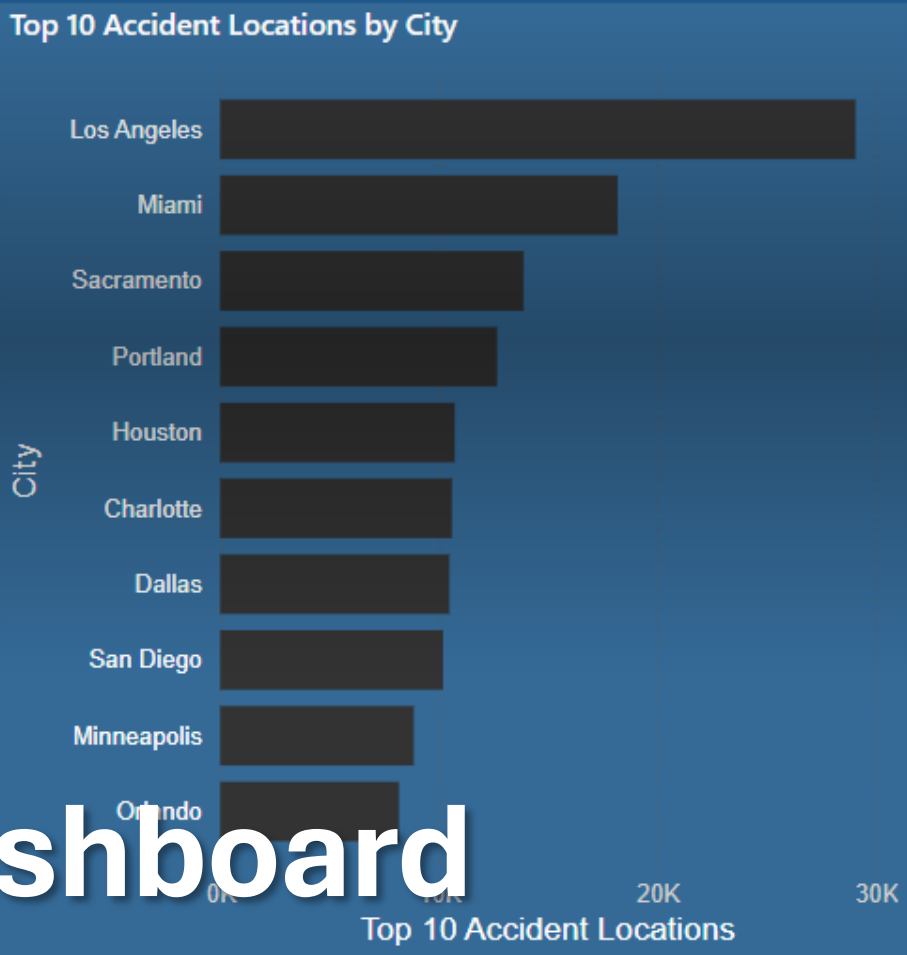
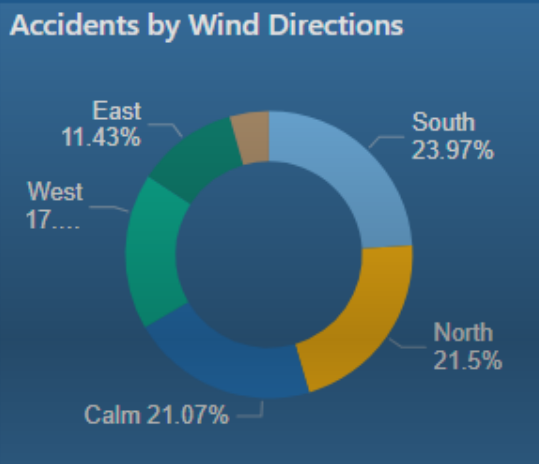
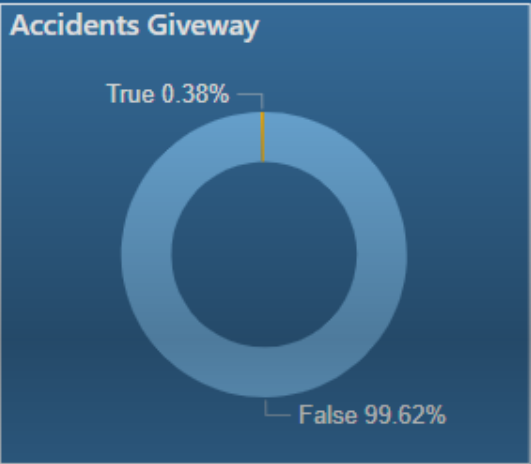
Very High Severe
80.9K
0.08

North East
100845

South
287137

South West
20707

West
591082



PowerBI Analysis Dashboard

Business Impact & ROI



Benefits

Considerable reduction
in response time

Cost savings in
emergency resources

Big improvement in
accident prediction
accuracy

Stakeholder Value

Law enforcement:
Optimal patrol routing

Emergency services:
Better resource
allocation

Insurance: Risk-
adjusted pricing

Public: Enhanced road
safety

Next Steps

Mobile app integration

Machine learning
model refinement

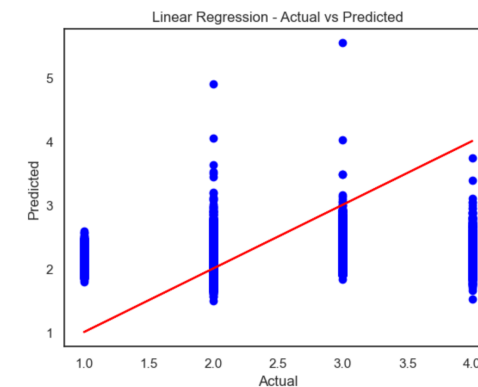
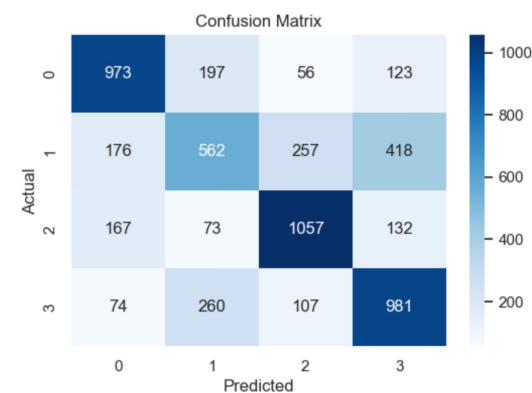
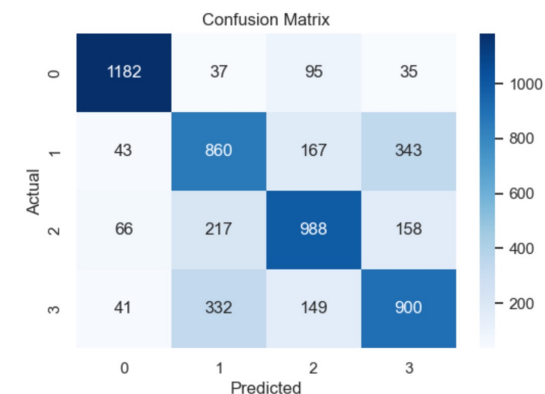
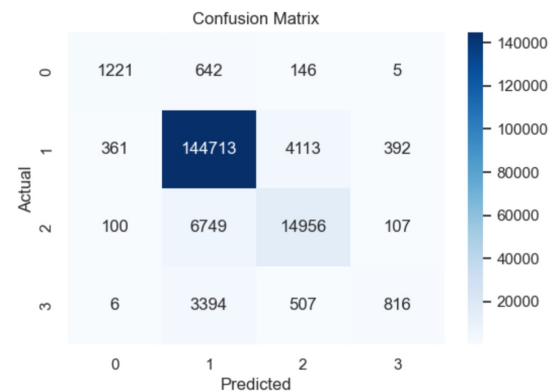
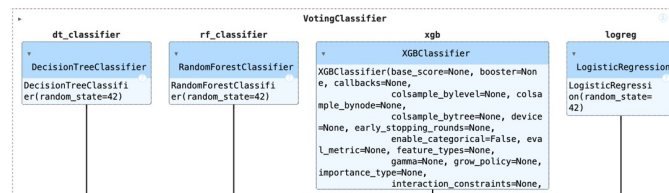
Real-time weather
data integration

Expansion to
additional cities

Model

- **Best Performer:** Random Forest (Highest test accuracy: 90.73%, F1 score: 89.91%).
- **Alternative:** Voting Classifier (Balanced and consistent performance).
- **Overfitting Issue:** Decision Trees show perfect training accuracy but poor generalization.
- **Baseline Model:** Logistic Regression serves as a simple benchmark.

Model	Train Score	Test Score	Precision	Recall	F1 Score	Accuracy
Voting Classifier	98.57%	89.79%	88.73%	89.79%	88.74%	89.79%
XGBoost	97.85%	78.21%	77.96%	78.21%	77.96%	78.21%
Random Forest	99.96%	90.73%	89.88%	90.73%	89.91%	90.73%
Decision Trees	100%	70.02%	70.16%	70.02%	70.08%	70.02%
Logistic Regression	63.40%	63.66%	63.02%	63.66%	63.05%	63.66%



Lessons Learned: Implementing BI for Accident Analytics

- **Data Management** – Data Quality & Preprocessing
- **Technical Implementation** -Process huge amount of data with performance optimization
- **Visualization** – Dashboard Implementation with better User Experience and with simple designs
- **Analytics Implementation** – Develop the model with higher accuracy and better performance

- **Recommendations**

- Short-term Actions**

- Enhanced patrol allocation in high-risk areas
 - Weather-based alert system implementation
 - Infrastructure improvement prioritization

- Long-term Strategies**

- AI-driven prediction model deployment with more additional features
 - Integrated emergency response system
 - Public awareness campaigns in high-risk areas